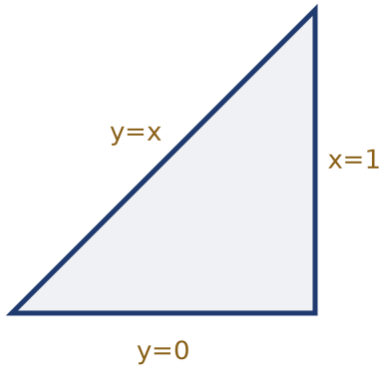
Double Integrals

ClipSAT Printable Practice Suite (clipsat.org) • Topic 15.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

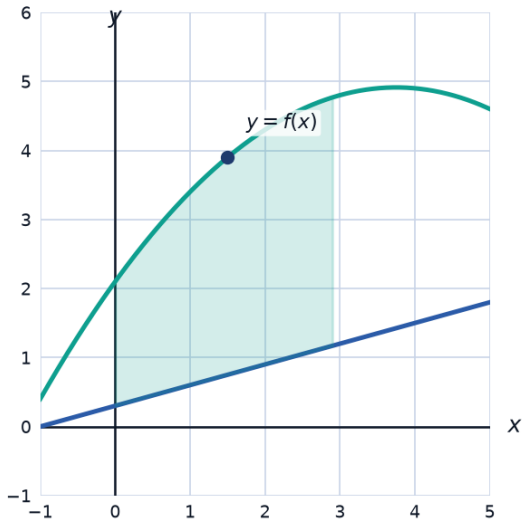
1 [Foundational] • Target: Double integrals over general regions

Evaluate $\int_0^1 \int_0^x (x + y)\, dy\, dx$, integrating over the triangular region $0 \le y \le x \le 1$.



2 [Foundational] • Target: Double integrals over general regions

Use a double integral to find the volume under $f(x, y) = 4 - x - y$ over the rectangle $0 \le x \le 1, 0 \le y \le 1$.



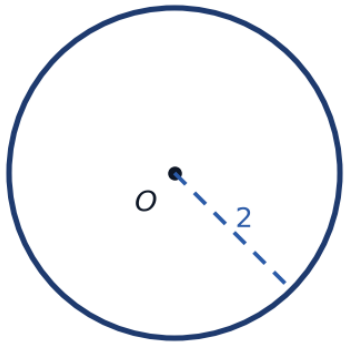
3 [Foundational] • Target: Reversing the order of integration

Evaluate $\int_0^1 \int_y^1 e^{x^2}\, dx\, dy$ by first reversing the order of integration (the region is $0 \le y \le x \le 1$).

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4 [Foundational] • Target: Double integrals in polar coordinates

Evaluate $\iint_D (x^2 + y^2)\,dA$ where D is the disk $x^2 + y^2 \le 4$, by converting to polar coordinates ($dA = r\,dr\,d\theta$).

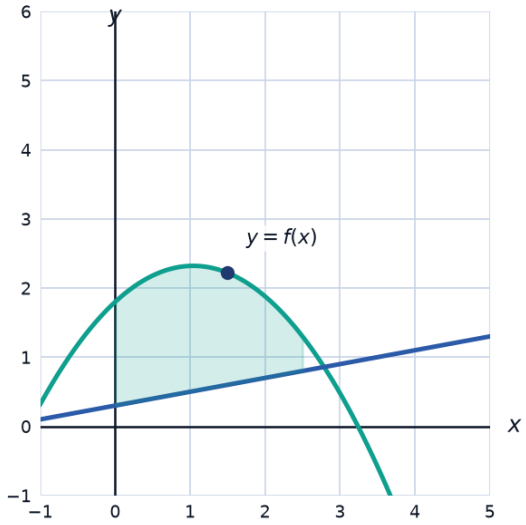


5 [Foundational] • Target: Double integrals in polar coordinates

Find the area of the region enclosed by $r = 2\cos\theta$ using a double integral in polar coordinates.

6 [Foundational] • Target: Average value over a region

Find the average value of $f(x, y) = xy$ over the rectangle $0 \le x \le 2$, $0 \le y \le 3$ (average value $= \frac{1}{A}\iint_D f\,dA$, where A is the region's area).



7 [Foundational] • Target: Mass and center of a region

A lamina occupies the rectangle $0 \le x \le 2$, $0 \le y \le 1$ with density $\rho(x, y) = x + y$. Find its total mass, $M = \iint_D \rho\,dA$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Mass and center of a region**

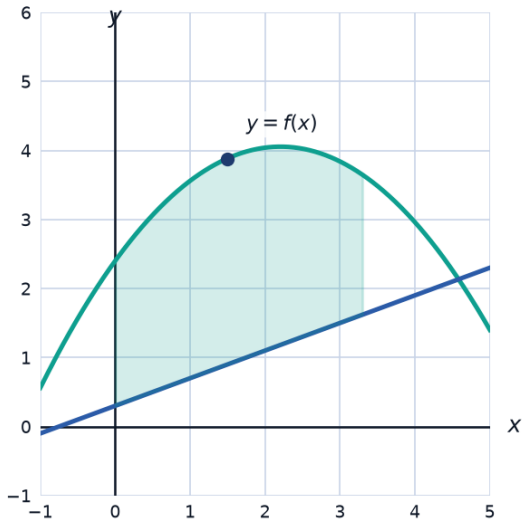
For a lamina occupying $0 \le x \le 1$, $0 \le y \le 1$ with constant density $\rho = 1$, find the x-coordinate of the center of mass, $\bar{x} = \frac{1}{M}\iint_D x\,dA$ (where M is the total mass).

9 **[Intermediate]** • **Target: Triple integral setup (a first look)**

Set up (do not evaluate) the triple integral for the volume of the solid bounded by $0 \le x \le 1$, $0 \le y \le 2$, $0 \le z \le 3$.

10 **[Intermediate]** • **Target: Double integrals with a variable upper limit**

Evaluate $\int_0^1 \int_0^{y^2} 3x\,dx\,dy$, integrating over the region $0 \le x \le y^2$, $0 \le y \le 1$.



11 **[Intermediate]** • **Target: Double integrals with a variable upper limit**

Set up and evaluate the double integral for the volume under $f(x, y) = xy^2$ over the triangular region bounded by $y = x$, $y = 0$, and $x = 2$.

12 **[Intermediate]** • **Target: An applied double-integral problem**

A metal plate occupies the region $0 \le x \le 3$, $0 \le y \le 2$, with temperature distribution $T(x, y) = 50 - x^2 - y^2$ (degrees Celsius). Find the average temperature over the plate.

13 **[Intermediate]** • **Target: Volume between two surfaces**

Find the volume of the solid between $z = 8 - x^2 - y^2$ and $z = 0$ over the region where $z \ge 0$ requires $x^2 + y^2 \le 8$, using polar coordinates.

14 **[Intermediate]** • **Target: Double integrals with exponential integrands**

Evaluate $\int_0^1 \int_0^1 xe^{xy}\,dy\,dx$ by integrating with respect to y first.

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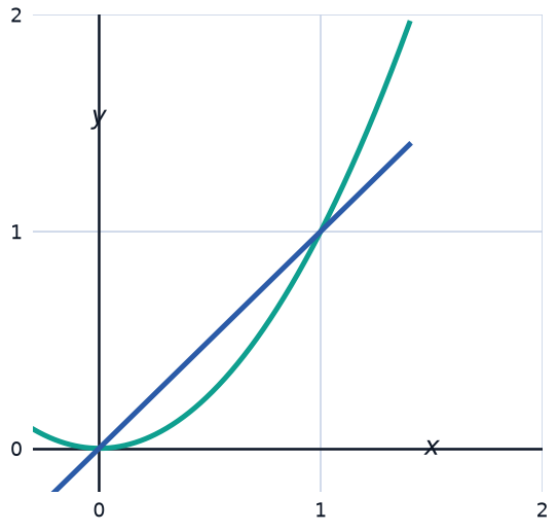
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: A capstone applied double-integral problem • Cross-Lesson: 8.03 Arc Length and Improper Integrals

An architect is designing a swimming pool whose floor depth varies according to $d(x, y) = 1 + 0.2x + 0.1y$ (meters) over a rectangular footprint $0 \le x \le 10$, $0 \le y \le 5$ (meters). Set up and evaluate the double integral for the total volume of water needed to fill the pool, then find the average depth by dividing by the pool's surface area.

16 [Challenging] • Target: A capstone applied double-integral problem • Cross-Lesson: 9.01 Separable Differential Equations

For the double integral $\iint_D (2x + y)\,dA$ where D is the region bounded by $y = x^2$ and $y = x$, complete each step:



a Find the intersection points of the two curves to determine the bounds.

b Set up the iterated integral with x as the outer variable.

c Evaluate the integral.

17 [Challenging] • Target: Average value over a region • Cross-Lesson: 1.01 Precalculus Review for Calculus

Level 2 · Medium Evaluate the iterated integral by reversing the order of integration: $\int_0^{\pi} \int_y^{\pi} \frac{\sin x}{x}\,dx\,dy$

18 [Challenging] • Target: Mass and center of a region • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Level 3 · Advanced Evaluate the double integral $\iint_D e^{x^2 + y^2}\,dA$ where D is the semi-annular region $\{(x, y) : 1 \le x^2 + y^2 \le 4, y \ge 0\}$.

19 [Challenging] • Target: Triple integral setup (a first look) • Cross-Lesson: 11.01 Sequences and Series

Level 2 · Medium Use the method of cylindrical shells to find the volume generated by revolving the region bounded by $y = 3x - x^2$ and $y = 0$ about the y-axis.

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20 [Challenging] • Target: Double integrals with a variable upper limit • Cross-Lesson: 12.01 Convergence Tests

Level 3 · AdvancedEvaluate the definite integral analytically: $\int_{-3}^{3}\left(x^5\sqrt{9-x^2}+\sqrt{9-x^2}\right)dx$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Iterated integrals over rectangles

Consider the solid E bounded by the surfaces $z=x^2+y^2$ and $z=8-x^2-y^2$. Calculate the volume of this solid using a triple integral by first transforming to cylindrical coordinates. Express your answer in terms of π.

a. $\frac{16\pi}{3}$

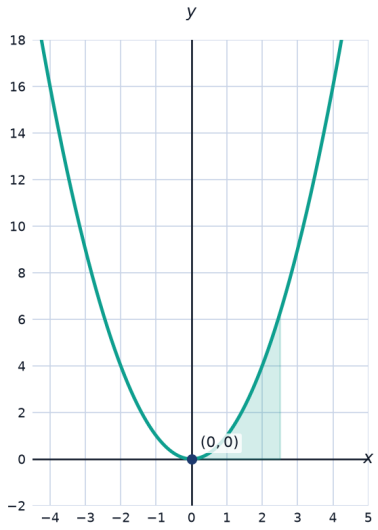
b. $\frac{32\pi}{3}$

c. 16π

d. $\frac{64\pi}{3}$

22 [Foundational] • Target: Double integrals over general regions

The figure shows the region R bounded by the curves $y=x^2$ and $y=4$. Which of the following double integrals represents the area of region R using the order of integration $dydx$?



a. $\int_{-2}^{2}\int_{x^2}^{4}dydx$

b. $\int_{-2}^{2}\int_{4}^{x^2}dydx$

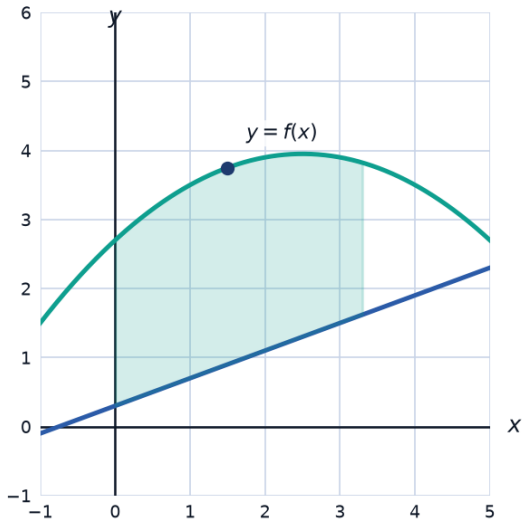
c. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}}dydx$

d. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}}dxdy$

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23 [Foundational] • Target: Reversing the order of integration

Iterated integral over tetrahedron $x + y + z \le 1, x, y, z \ge 0$: volume = 1/6. The integral $\iiint x\,dV =$



a. 1/6

b. 1/24

c. 1/12

d. 1/8

24 [Foundational] • Target: Double integrals in polar coordinates

The region between $y = x$ and $y = x^3$ in the first quadrant is revolved about the y-axis. Find the volume.

a. $\frac{4\pi}{21}$

b. $\frac{4\pi}{15}$

c. $\frac{2\pi}{15}$

d. $\frac{4}{15}$

25 [Foundational] • Target: Average value over a region

The double integral over region D of $f = 1$ gives:

a. area of D

b. perimeter of D

c. volume

d. 0

26 [Foundational] • Target: Mass and center of a region

Evaluate $\int_0^2 \int_y^2 e^{x^2}\,dx\,dy$ by reversing the order of integration.

a. $-\frac{1}{2} + \frac{e^4}{2}$

b. $-1 + e^4$

c. $\frac{1}{2} + \frac{e^4}{2}$

d. $-\frac{1}{4} + \frac{e^4}{4}$

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27 [Foundational] • Target: Triple integral setup (a first look)

The volume of the region $1 \le x^2 + y^2 + z^2 \le 4$ (shell between spheres of radii 1 and 2):

a. $4\pi/3$

b. 4π

c. 8π

d. $4\pi(8-1)/3 = 28\pi/3$

28 [Foundational] • Target: Double integrals with a variable upper limit

Use spherical coordinates to find the volume of the first-octant part of the ball $x^2 + y^2 + z^2 \le 16$.

a. $\frac{256\pi}{3}$

b. $\frac{64\pi}{3}$

c. $\frac{32\pi}{3}$

d. $\frac{16\pi}{3}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: An applied double-integral problem

Volume under $z = 1$ over region R of area A is:

a. A^2

b. A

c. $A/2$

d. πA

30 [Intermediate] • Target: Volume between two surfaces

Use polar coordinates to evaluate $\iint_R xy\,dA$, where R is the part of the disk $x^2 + y^2 \le 16$ in the first quadrant.

a. 32

b. 32π

c. 64

d. 16

31 [Intermediate] • Target: Double integrals with exponential integrands

Volume between $z = x^2 + y^2$ (paraboloid) and $z = 1$ over disk $r \le 1$:

a. $3\pi/4$

b. π

c. $\pi/4$

d. $\pi/2$

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32 [Intermediate] • Target: A capstone applied double-integral problem

Volume between $z = 1$ and $z = x^2 + y^2$ over unit disk:

a. $\pi/4$

b. $\pi/2$

c. π

d. $3\pi/4$

33 [Intermediate] • Target: Iterated integrals over rectangles

Use a double integral to find the area of the region under $y = x^2$ for $0 \le x \le 1$.

a. $\frac{2}{3}$

b. $\frac{1}{3}$

c. $\frac{1}{6}$

d. $\frac{1}{2}$

34 [Intermediate] • Target: Double integrals over general regions

Volume between planes $z = 0$ and $z = 3x + 2y$ over $D = [0, 1] \times [0, 1]$:

a. 5/2

b. 3

c. 1

d. $3/2 + 1 = 5/2$

35 [Intermediate] • Target: Reversing the order of integration

Volume integral: using spherical coordinates, $dV =$

a. $dr\,d\theta\,d\phi$

b. $r^2 \sin\phi\,dr\,d\theta\,d\phi$

c. $r\sin\theta\,dr\,d\theta\,d\phi$

d. $r^2\,dr\,d\theta\,d\phi$

36 [Intermediate] • Target: Double integrals in polar coordinates

Volume under $z = f(x, y)$ over region D equals:

a. $\iint_D f^2\,dA$

b. $\oint_D f\,ds$

c. $\iint_D ablaf\,dA$

d. $\iint_D f\,dA$

37 [Intermediate] • Target: Average value over a region

Volume between paraboloid $z = x^2 + y^2$ and $z = 1$ over unit disk:

a. $3\pi/4$

b. π

c. $\pi/4$

d. $\pi/2$

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Mass and center of a region • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

For iterated integral $\int_a^b \left(\int_{g_1(x)}^{g_2(x)} f(x,y)dy \right) dx$: the inner integral is done first, then the outer. True or false?

a. False — outer first

b. True

c. Only for separable functions

d. Depends on Fubini

39 [Challenging] • Target: Triple integral setup (a first look) • Cross-Lesson: 11.01 Sequences and Series

Consider a region in the first quadrant bounded by the curve $y = 9 - x^2$, the x-axis, and the y-axis. A rectangle is inscribed within this region such that one vertex is at the origin, two sides lie along the x and y axes, and the fourth vertex lies on the curve $y = 9 - x^2$. If the area of this inscribed rectangle is maximized, determine the value of the definite integral that represents the volume of a solid formed by revolving the region bounded by the curve $y = 9 - x^2$, the x-axis, and the y-axis about the y-axis.

a. $\frac{81\pi}{2}$

b. 81π

c. 162π

d. $\frac{243\pi}{2}$

40 [Challenging] • Target: Double integrals with a variable upper limit • Cross-Lesson: 12.01 Convergence Tests

Consider a region R in the first quadrant bounded by the curve $y = e^{-x^2}$, the x-axis, and the y-axis. Let S be the solid generated when R is revolved about the y-axis. The volume of S can be calculated using the method of cylindrical shells or the disk/washer method. However, direct application of standard integration techniques to find the volume of S leads to an integral that cannot be evaluated in terms of elementary functions. A related problem involves finding the surface area of a solid of revolution. Let C be the curve defined by $y = \sqrt{1 - x^2}$ for $0 \le x \le 1$. This curve represents a quarter-circle. If this curve C is revolved about the x-axis, it generates a hemisphere. The surface area of this hemisphere (excluding the base) is given by the integral $A = \int_0^1 2\pi y \sqrt{1 + (y')^2}\, dx$. Determine the exact surface area of the generated hemisphere.

a. $\frac{\pi}{2}$

b. π

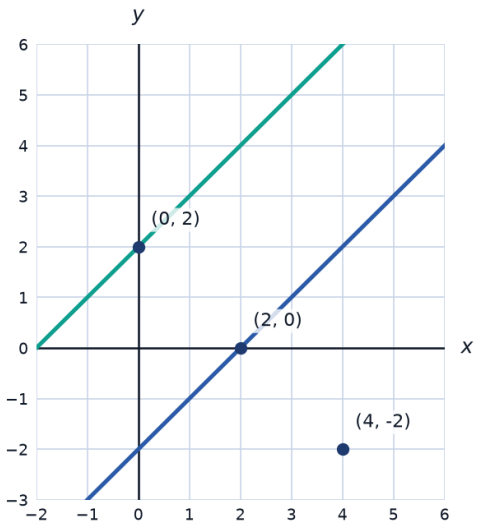
c. 2π

d. 4π

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41 [Challenging] • Target: An applied double-integral problem • Cross-Lesson: 12.02 Power Series and Taylor Series

The figure shows the graph of a continuous function $f(x)$ on the interval $[a, d]$. The shaded region represents the definite integral $\int_a^c f(x)\,dx$. If the area of the shaded region is 12 square units and the area of the unshaded region between $x = c$ and $x = d$ is 5 square units, what is the value of $\int_a^d f(x)\,dx$?



a. 17

b. 7

c. 2.4

d. -7

42 [Challenging] • Target: Volume between two surfaces • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Volume of region bounded by $x^2/4 + y^2/9 = 1$ rotated about the x-axis:

a. 24π

b. 12π

c. 48π

d. 36π

43 [Challenging] • Target: Double integrals with exponential integrands • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Region $x \geq 0$, $y \geq 0$, $x + y \leq 1$: volume when rotated about y-axis by shell method:

a. $\frac{\pi}{3}$

b. $\frac{\pi}{4}$

c. $\frac{\pi}{2}$

d. $\frac{\pi}{6}$

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44 [Challenging] • Target: A capstone applied double-integral problem • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Pappus's theorem: volume of solid of revolution = $2\pi\bar{x} \cdot A$ where $\bar{x}$ is the centroid of the region and A is the:

a. arc length

b. surface area

c. area of the region

d. cross-sectional area

45 [Challenging] • Target: Iterated integrals over rectangles • Cross-Lesson: 2.01 Evaluating Limits

Volume of region $y \geq x^2$, $y \leq 4$ rotated about $y = 4$ using washers:

a. $256\pi/15$

b. $128\pi/15$

c. $512\pi/15$

d. $64\pi/15$

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